

Master Compliance Manual: Residential Floor Plan Design

Based on Bangalore Mahanagara Palike Building Bye-Laws 2003

This document serves as a definitive checklist and design guide for architects and planners creating residential floor plans in Bangalore. It isolates the geometric and spatial rules required for plan approval.

1. Site Planning & Building Envelope

Before detailing the interior, the allowable building footprint must be defined by applying setbacks, coverage limits, and clearance rules.

1.1 Electrical Line Clearances (Mandatory Exclusion Zones)

No building construction is permitted below electrical lines. The floor plan must respect these horizontal safety buffers:

- **Low/Medium Voltage Lines:** Minimum **1.2m** horizontal clearance.
- **High Voltage (up to 11 KV):** Minimum **1.2m** horizontal clearance.
- **High Voltage (11 KV - 33 KV):** Minimum **2.0m** horizontal clearance.
- *Note:* Vertical clearance (2.5m - 3.7m) must also be considered in the section/elevation.¹

1.2 Setbacks (Exterior Open Spaces)

The building footprint is carved out of the plot by mandatory open spaces.

A. Low-Rise Buildings (Height ≤ 9.5m)

Setbacks are determined by the Depth of the site (for Front/Rear) and the Width of the site (for Sides).

Site Dimension (Depth or Width)	Front Setback (Min)	Rear Setback (Min)	Left Setback (Min)	Right Setback (Min)
Up to 6.0m	1.00m	1.00m	0m / 1.00m*	1.00m
Over 6.0m up to 9.0m	1.00m	1.00m	1.00m	1.00m

Over 9.0m up to 12.0m	1.50m	1.50m	1.50m	1.50m
Over 12.0m up to 18.0m	1.50m	3.00m	1.50m	3.00m
Over 18.0m up to 24.0m	4.00m	3.00m	2.50m	3.50m
Over 24.0m	3.50m	5.00m	3.00m	4.00m

Note: For corner plots, both sides facing the road are treated as "Front".¹

B. High-Rise Buildings (Height > 9.5m)

For taller buildings, a uniform setback applies to all four sides.

Building Height	Minimum Setback on All Sides
9.5m - 12.0m	4.5m
12.0m - 15.0m	5.0m
15.0m - 18.0m	6.0m
18.0m - 21.0m	7.0m
21.0m - 24.0m	8.0m
Above 50m	16.0m
¹	

1.3 Permitted Projections in Setbacks

- **Cantilever (Chajja):** Max projection **0.75m** or 1/3rd of setback width (whichever is less).
- **Balconies:** Permitted in setbacks (except Ground Floor). Max projection **1.1m** (1st Floor) to **1.75m** (Upper Floors), but never more than 1/3rd of the setback width.
 - *Constraint:* Total length of balconies cannot exceed 1/3rd of the building length on that side.
- **Cantilever Portico (Car Porch):** Allowed in side setback on Ground Floor.

- Max Size: **3.0m width x 4.5m length**.
- Clearance: Must leave 1.0m open from the property boundary.
- Condition: Must be open on sides; terrace not accessible.¹

2. Internal Spatial Standards (Room Dimensions)

The floor plan must label dimensions that meet or exceed these minima.

2.1 Habitable Rooms (Living, Bed, Dining, Study)

- **Standard Minimum Area: 9.5 sq.m.**
- **Small Plot Exception:** If plot $\leq$ 120 sq.m, minimum room area is **8.0 sq.m.**
- **Minimum Width: 2.4m** (One dimension must be at least this wide).
- **Minimum Height: 2.75m** (2.4m allowed for AC rooms).
- **Plinth Height:** Ground floor level must be at least **0.45m** above ground level.¹

2.2 Kitchens

- **Kitchen (Standard):** Min Area **5.0 sq.m**; Min Width **1.8m**.
- **Kitchen (with separate Store):** Min Area **4.5 sq.m**.
- **Kitchen-cum-Dining:** Min Area **7.5 sq.m**; Min Width **2.1m**.
- **Requirement:** Must show position of flue/chimney.¹

2.3 Bathrooms & Water Closets (WC)

- **Bathroom (Separate):** Area **1.8 sq.m**; Width **1.2m**.
- **WC (Separate):** Area **1.1 sq.m**; Width **0.9m**.
- **Combined Bath & WC:** Area **2.0 sq.m**; Width **1.2m**.
- **Height:** Min **2.2m** (allows for overhead lofts).¹

2.4 Intermediate Floors

- **Mezzanine:**
 - Allowed between Ground and First floor only.
 - Max Area: 1/3rd of Ground Floor covered area.
 - Min Height: **2.2m**.
- **Lofts (Storage):**
 - Permitted above Kitchens (max 25% coverage), Baths/Corridors (100% coverage).
 - Max Headroom: **1.5m**.¹

3. Light and Ventilation

Every habitable room must have openings (windows/ventilators) opening directly to the

exterior or an interior open courtyard.

3.1 Aggregate Area of Openings

- **Residential Requirement:** Window area must be at least **1/6th** of the floor area of the room.
- **Kitchens:** Window area increased by **25%** over the standard requirement.
- **Depth Rule:** No part of a room is considered "lighted" if it is more than **7.5m** from an opening.¹

3.2 Ventilation Shafts (Internal Toilets)

If a Toilet/WC does not open to the exterior, it must open into a ventilation shaft of minimum dimensions:

No. of Storeys	Minimum Shaft Area	Minimum Width
Up to 2	1.2 sq.m	0.9m
Up to 4	1.8 sq.m	1.2m
Up to 6	4.0 sq.m	1.5m
11 and above	9.0 sq.m	3.0m
¹		

4. Circulation: Stairs, Lifts, and Exits

The core circulation areas are strictly regulated for safety.

4.1 Staircases

- **Minimum Width:**
 - Residential Dwellings (General): **1.0m**.
 - Row Housing (2 storeys): **0.75m**.
 - Apartments / High Rises: **1.5m**.
- **Tread (Go):** Min **25 cm** (Residential).
- **Riser:** Max **19 cm** (Residential).
- **Flights:** Max 15 risers per flight.
- **Headroom:** Min **2.2m** measured vertically from nosing.

- **Fire Escape:** Required for buildings >15m height (External metal/concrete stair).¹

4.2 Corridors and Exits

- **Door Widths:**
 - Main Exits: Min **1.0m** (Non-residential) / **0.9m** (Residential suggested, usually main doors are 1.0m+).
 - Toilet Doors: Standard practice ~0.75m (Plan check: "Exit doorways shall not be less than 75cm").
- **Travel Distance:** Max distance from any point in the unit to a staircase/exit is **22.5m** (Residential).
- **Exits:** Two exits required per floor for buildings >15m height.¹

4.3 Lifts

- **Mandatory:** For buildings >15m height OR > G+3 floors.
- **Capacity:** For high rises, one lift must accommodate a wheelchair (min internal car size approx 1.1m x 2.0m).¹

5. Parking Standards

Parking must be integrated into the floor plan (Stilt or Basement).

5.1 Parking Slot Dimensions

- **Car:** 3.0m x 6.0m (18 sq.m).
- **Two-Wheeler:** 1.25m x 2.0m (2.5 sq.m).
- **Cycle:** 0.75m x 2.0m (1.5 sq.m).

5.2 Driveways & Ramps

- **Driveway Width:** Min 2.5m.
- **Ramp to Basement:**
 - Min Width: **3.5m**.
 - Slope: Max **1:10**.
 - Start Point: Must leave a 2.0m buffer inside the boundary before the ramp starts.¹

5.3 Quantum (Residential)

- **Single Family:** 1 car space for 100-150 sq.m unit size.
- **Apartments:**
 - Units <50 sq.m: 1 car per 2 units.
 - Units 50-150 sq.m: 1 car per unit.
 - **Visitor Parking:** Add **10%** extra for visitors.¹

6. Service Areas (Basement & Utility)

6.1 Basement Regulations

- **Permitted Use:** Parking, AC Plants, Electrical Panels, Strong Rooms. (Strictly **NO** Habitable use).
- **Height:** Min **2.4m**; Max **2.75m** (can be 3.6m for mechanical parking/machinery).
- **Projection:** Roof cannot project more than **1.0m** above ground level.¹

6.2 Other Services

- **Refuse Chutes:** Mandatory for High-Rise Residential. Must be located away from the staircase/lift core.
- **Mail Boxes:** Must be provided on the Ground Floor for multi-storey buildings.
- **Rainwater Harvesting:** Plan must show location of recharge pit/tank (Min 200 sq.m plot).
- **Solar Water Heaters:** Mandatory plumbing duct for solar connection on terrace (Min 200 sq.m plot).¹

7. Typology-Specific Constraints

7.1 Row Housing (Min 3 units)

- **Min Plot:** 210 sq.m combined.
- **Setbacks:** Front 2.0m, Rear 1.5m. **Side setbacks not required** between units, only at the ends of the block.¹

7.2 Accessibility (Schedule XI)

For apartment buildings (Public/Semi-public nature):

- **Ramp:** Entrance must have a ramp (1:12 slope, 1.8m width).
- **Lift:** One wheelchair-accessible lift.
- **Toilet:** One accessible toilet on Ground Floor (**1.5m x 1.75m**).¹

8. FAR & Floor Area Calculation

- **Included in FAR:** All floor area, walls, lobbies, corridors.
- **Excluded from FAR (The "Free" Areas):**
 - Parking areas (Stilt/Basement).
 - Staircase rooms & Lift shafts.
 - Ducts (Sanitary/Electrical).

- Balconies (Open).
- Water tanks & Pump rooms.
- Terrace structures (Lift machine room, Stair head).
- *Note:* A penthouse (up to 19 sq.m) is exempted from FAR.¹

Works cited

1. Bangalore-Building-Byelaws (2).pdf